Tyler Beach

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Experience

Firmware Validation Engineer February 2026 - Present
Solidigm *Vancouver, BC*

- Validate **NVMe transport layer** mechanics and read/write protocol compliance over **PCIe interfaces**, debugging hardware-firmware boundary conditions and low-level command queues.
- Architected an automated test suite verifying strict compliance with **OCP (Open Compute Project) Datacenter SSD specifications** to validate hardware-firmware boundaries on physical post-silicon platforms.
- Utilize **Python** to author modular **black-box testing** modules within a proprietary framework, significantly expanding transport-layer test coverage and accelerating regression detection cycles.

Software Developer Intern January 2025 - April 2025
QOL MedTech *Calgary, AB*

- Architected and launched a scalable, multi-tenant web app for a fast-paced startup using **Next.js, Fastify, TypeScript, and Tailwind CSS**, enabling remote physiotherapy for clients, trainers, and admins.
- Built and documented RESTful APIs in **Fastify** and integrated them with a **PostgreSQL** database, enabling reliable data flow between the frontend and backend services.
- Designed and implemented secure, role-based authentication and authorization workflows using **JWT**, **AWS Cognito**, **RDS**, and **S3**, ensuring HIPAA-compliant data access and storage.

Projects

Gazprea Compiler | *C++, LLVM, MLIR, ANTLR4* Website

- Developed a compiler for a general-purpose language, generating **LLVM IR** using **MLIR** and supporting features like function calls, loops, conditionals, type aliasing, vectors and 2D arrays.
- Implemented a lexer and parser using **ANTLR4**, generating a custom heterogeneous Abstract Syntax Tree (AST).
- Wrote **2000+ unit tests** for every feature, ensuring comprehensive coverage and reliability, testing against a competitive suite of 600+ test cases, maintaining quality and correctness.
- Achieved **1st overall ranking among 12 teams**, outperforming peers in runtime performance, memory management, and feature completion.

ASCII / Voxel Renderer | *C++, SDL2* Website

- Built a real-time software 3D renderer from scratch in C++, progressing from ASCII ray-casted worlds to a fully **voxel-based engine** with lighting and shadows.
- Optimized performance through algorithmic improvements (**DDA vs ray marching**), buffering strategies, and incremental rendering refinements.
- Ported the renderer to **SDL2**, rewriting core systems to support windowing, input handling, double buffering, minimap visualization, and smooth real-time interaction.

MediMinutes | *Next.js, TypeScript, Tailwind, AWS, PostgreSQL, Prisma, Python* Website

- Engineered a gamified medical learning platform featuring **daily crosswords**, **symptom-diagnosis matching**, and **interactive quizzes**, with 200+ topics and progress tracking via personalized dashboards.
- Created an automated content pipeline using **Beautiful Soup** and **custom algorithms** to generate daily puzzles from 20k+ medical data points, eliminating manual updates.
- Implemented secure authentication (**Google OAuth + custom auth**) and real-time leaderboards to foster competitive learning among medical students.

Education

University of Alberta December 2025
Bachelor of Science in Computer Science *Edmonton, AB*

Skills

Languages: Python, C++, C, TypeScript, RISC-V Assembly, SQL
Technologies: Pytest, GitHub Actions CI/CD, Docker, AWS, PostgreSQL, ANTLR4, LLVM/MLIR
Coursework: Compiler Design, Computer Architecture, Operating Systems, Software Engineering, Computer Networking, Image Processing, Reinforcement Learning